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Glycome-Proteome Interactome Cartography via Proximity Tagging

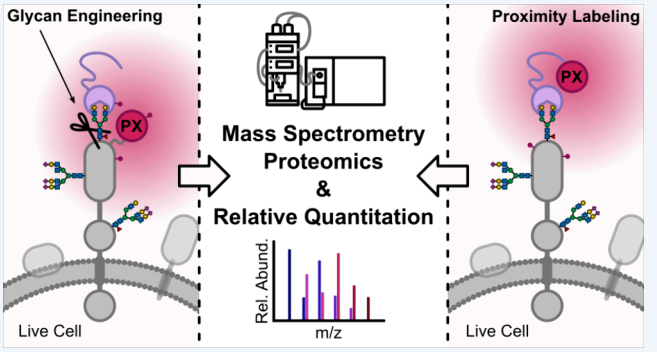
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CONSPECTUS: Glycans are now recognized as essential biomolecules for life, and an increasing number of investigators and studies continue to reveal the myriad ways in which these post-translational modifications regulate important biological events. Chief among the ways in which glycans carry out their roles is by engaging in binding interactions with glycan-binding proteins (GBPs). Such interactions are important for proper physiology or in the activation of disease. Thus, achieving a precise molecular understanding of these interactions can pave new avenues for synthetic control and potentially therapeutic strategies to regulate disease. In this Account, we discuss our efforts toward revealing the collective interactions between glycans, protein glycoconjugates, and GBPs in cells, with an eye toward identifying the specific glycan-carrying proteins that interact with the GBPs in a functional manner.



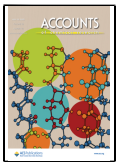
While the importance of studying glycan-GBP interactions has long been established, prior to our work, much of glycoscience had been occupied with the systematic assignment of the structural features of glycans required for recognition by GBPs in vitro, often using homogeneous glycan arrays. This important body of work enabled the preliminary identification of principal GBP-glycan binding preferences and allowed the rational design of functionalized glycan molecules to use as competitive inhibitors. Equipped with these two sets of important tools, expanding our understanding of glycan-GBP interactions from in vitro glycan binding preferences toward that of glycoprotein-GBP interactions *in vivo* has become timely and possible. We describe the conceptual motivation behind developing proximity tagging technologies to capture these glycoprotein-GBP interactions and how we used these interactome data sets to guide the discovery of functional interactors or create new hypotheses. Furthermore, we describe how proximity tagging can continuously be used toward application in our work, and the potential routes of investigation it may facilitate in the coming years.

- Joeh, E.; O'Leary, T.; Li, W.; Hawkins, R.; Hung, J. R.; Parker, C. G.; Huang, M. L. Mapping glycan-mediated galectin-3 interactions by live cell proximity labeling. *Proc. Natl. Acad. Sci. U. S. A.* **2020**, *117*: 27329−27338.[1] *This study showed that appending proximity tagging enzymes, such as APEX2, to glycan-binding proteins, could be used as a general strategy to capture glycan-protein interactions in live cells.*
- O'Leary, T. O.; Critcher, M.; Stephenson, T. N.; Yang, X.; Hassan, A.; Bartfield, N. M.; Hawkins, R., Huang, M. L. Chemical editing of proteoglycan architecture. *Nat. Chem. Biol.* **2022**, *18*: 634−642..[2] *This study showed how various chemical biology techniques could be integrated to generate glycosite- and glycoform- controlled proteoglycan glycoconjugates, and it showed how proteoglycan glycosylation states could alter interactomes using proximity tagging.*
- Hassan, A. A.; Wozniak, J. M.; Vilen, Z.; Li, W.; Jadhav, A.; Parker, C. G.; Huang, M. L. Chemoproteomic mapping of human milk oligosaccharides (HMO) interactions in cells. *RSC Chem. Biol.* **2022**, 3: 1369−1374..[3] *While a large majority of glycans present in cells are glycoconjugates, a fraction exists as soluble unconjugated molecules, too small to append a macromolecular enzyme. This manuscript demonstrates the use of photo-cross-linking strategies to capture the interactions of free glycans.*

INTRODUCTION

More than half of the proteome and >90% of the cell surface proteome are modified with glycan post-translational mod-

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ifications (PTMs).[4] In addition to basic structural roles like enhancing protein stability[5−7] or modulating protein conformation,[7−9] glycans can also play important roles in facilitating biological functions, like immune signaling,[10−14] cellular differentiation,[15,16] and cell adhesion.[17−20] Many of these functional consequences are initiated by binding interactions between glycans on carrier proteins and glycan binding proteins (GBPs) at the cell surface (Figure 1).[21] Although GBPs can

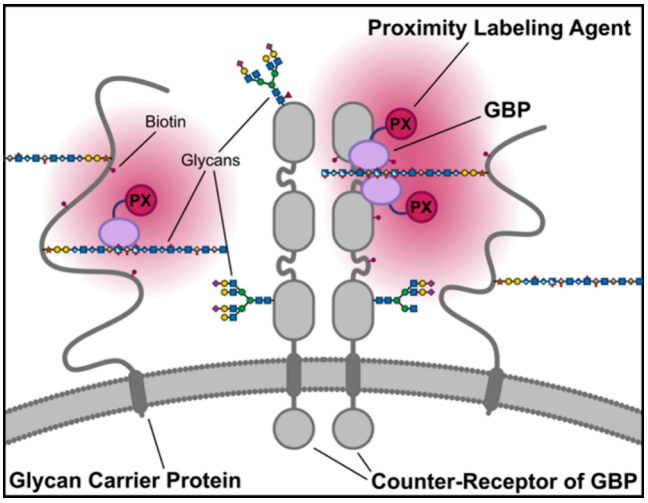


Figure 1. Schematic diagram of a glycoprotein-glycan binding protein (GBP) interaction. Proximity tagging agents appended to GBPs facilitate biotinylation of the interactome. PX indicates the proximity tagging ascorbate peroxidase enzyme APEX2.

interact with other types of conjugates, such as glycolipids or glycoRNA, we will limit our focus in this Account toward interactions between GBPs and protein-based glycoconjugates.

While there is already a rich body of information on the biological functions of GBPs and certain glycoproteins, matching the appropriate pair in the correct tissue, cell type, and function, is not always an easy endeavor. Furthermore, current techniques to study GBP interactions often affect the glycome as a whole, limiting the conclusions to such experiments, and often implying that similarly glycosylated protein play redundant roles. Contrarily, several lines of evidence suggest that there might be more specificity to these glycan-GBP interactions beyond simple recruitment. First, genome-wide screens for glycoproteins required for GBP function suggest that only select subsets of proteins among the thousands of glycoproteins are required for function.[22,23] Second, there is an accumulating body of work showing the sensitivity of glycan binders to modifications on the glycan carrier protein.[24−27] While we do not yet have a comprehensive understanding of all the factors regulating glycosylation, and in turn, we cannot yet predict which specific glycoproteins interact with GBPs for function, we can use *"binding-first"* approaches to empirically determine the interactors for GBPs and glycoproteins and match these partners together.

CHALLENGES OF STUDYING GLYCAN-GBP INTERACTIONS

The first challenge in the study of glycan-GBP interactions involves the heterogeneous nature of glycan modifications arising from their biosynthesis. Glycans are installed onto a carrier protein by a sequential series of tens to hundreds of enzyme-driven steps in the endoplasmic reticulum and Golgi apparatus, conferring high chemical heterogeneity.[28] Consequently, most protein glycoconjugates exhibit variability in both glycosylation site (glycosite) occupancy and the structural composition of the glycan.[29−31] Thus, any glycoprotein exists in a multitude of glycosylation variants (glycoforms) that are difficult to predict with high certainty and specificity. Certain factors are known to influence glycosylation status. At the molecular scale, glycosylation is affected by the residence time of a protein along the glycan biosynthetic pathways, expression of various glycosyltransferases, and nucleotide sugar levels.[28,32,33] At the cellular scale, glycosylation is affected by changes in cell state[34−36] arising from shifts in cellular metabolism[37] or stress.[38] Combined, these factors increase the complexity of measuring protein glycoforms as well as the glycan-GBP interactions that result from them.

Second, the conformational flexibility and hydrophilicity of glycans present a unique problem in their interactions with proteins, often resulting in binding with low thermodynamic affinities on a monomeric basis ($K_D \sim$ mM).[39] The biological solution to this poor binding is extreme reliance on the advantages of avidity, with most GBPs either possessing multiple binding sites or the capability to oligomerize.[39−41] This feature sensitizes GBPs to minute differences in glycan structure, the structure of the carrier protein carrying the glycans, and the overall presentation of these partners within the fluid organization of membrane microdomains. In the context of physiology and biological function, GBPs and glycoproteins make for interesting targets to study, but these facets also increase the difficulty and stringency required in recapitulating physiologically relevant conditions when studying their interactions.[42] These characteristics also render the identification of glycan-GBP interactions largely incompatible with traditional high-throughput affinity-based identification techniques using cell lysates, leading to a lack of information on glycoprotein-GBP interactions within well-known protein affinity databases (e.g., BioPlex, BioGRID).

Consequently, even if a GBP may bind to a glycan in isolation (i.e., in vitro), it does not necessarily imply that there will be the same binding interaction or associated response in physiological contexts. The simplistic model that a glycan motif on its own offers enough structural diversity and information to define the effects of a given GBP can be true for some GBPs, like the Ashwell-Morell receptor and its associated hepatic clearance of desialylated glycoproteins in blood.[43,44] However, many more functionally diverse outcomes can occur due to the properties and identity of the carrier protein. A well-characterized example of this is in the binding between P-selectin and PSGL-1, mediating leukocyte rolling, where P-selectin is known to bind sialyl Lewis^X, but it also requires the sulfation of a specific tyrosine residue present on PSGL-1.[17,24,45] For a holistic understanding of GBP function, it is important to attempt to capture these glycan-GBP interactions where the binding occurs in its most endogenous and physiologically relevant state.

In the absence of a general unifying principle for how glycosylation alters protein structure and function, computational approaches currently offer little to predict glycan interactions and functions. And so, despite advances in proteomics and glycomics expanding the capacity to identify glycans, the hypothesis-driven study of functional glycans in cellular contexts often relies on a researcher's intuition. To remedy this deficiency, target-agnostic approaches that "fish" or "screen" for GBP counter-receptors have become the primary

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strategy for elucidating novel functions arising from glycan-GBP interactions.

BRINGING PROXIMITY TAGGING INTO THE STUDY OF GLYCOPROTEIN-GBP INTERACTIONS

Motivated by the need to capture glycoprotein-GBP complexes in cells, we sought to apply techniques often used in the identification of protein−protein interactions (PPIs), as the characterization of these interactions face much of the same considerations and challenges. The core analytical challenge of studying protein interactions lies in sifting through the roughly ∼ 30,000 proteins[46] present in a cellular proteome and identifying interactors while minimizing false positives. Even the most effective techniques will find false-positive results, so in-depth validation of the results is necessary to verify whether the PPIs identified occur. In a sense, the initial searches for novel PPIs are focused on the reduction of search space from the thousands of proteins present in a cellular proteome, to a handful of promising results.

Historically, there have been few target-agnostic methods for identification of novel PPIs. Two notable examples are the yeast two-hybrid (Y2H)[47] and affinity purification mass spectrometry (AP-MS)[48] techniques, which were first reported in 1989 and 1999, respectively. Y2H identified PPIs by engineering the bait protein and the protein of interest (POI) into yeast, tagging each protein with either the DNA binding domain or activation domain of a transcriptional activator. The ability of the bait and POI to induce the proximity of the two functional domains was read out by the transcription of the gene into a reporter, which could result in luminescence, fluorescence, or production of a colorimetric substrate depending on the reporter gene. AP-MS on the other hand, takes advantage of the advancements in mass spectrometry proteomics, analyzing the complex mixture of proteins pulled down by coimmunoprecipitation using mass spectrometry.

For identifying glycoprotein-GBP interactions, both Y2H and AP-MS suffer from reliance on strong binding interactions and identification of interactors in non-native cell lysates, which were critical parameters when considering the interactomes of GBPs. We also needed to ensure that if we were to modify the GBP or counter-receptors with a tag, that tag would minimally interfere with the natural oligomerization of the GBP or distribution of the counter-receptors. Affecting either the valency of the GBP or the cellular presentation of counter-receptors could likely introduce unnatural, physiologically irrelevant interactions. At the same time, the technique would also need to be able to rapidly capture interactions, because glycan-GBP interactions can be dynamic and transient. In addition, the ability for the technique to function intracellularly as well as extracellularly in both reducing and oxidative environments would also be very useful for tracking changing interactomes as complexes move into cells.

When combining all these considerations, it was quite apparent that the Y2H and AP-MS approaches would intrinsically be incapable of studying glycan-GBP interactions in a live-cell experiment with minimal perturbations to the carrier proteins or GBPs. Other techniques like photo-cross-linking have demonstrated utility in capturing such transient interactions; however, these methods were better used for studying small molecule binding, as with metabolically engineered glycans[49] or with free oligosaccharides, as we have demonstrated with human milk oligosaccharides.[3]

Ultimately, it was the growing field of proximity tagging that seemed to fulfill all the requirements necessary for the study of glycoprotein-GBP interactions. Proximity tagging enabled the *in situ* detection of protein proximity in live cells, facilitating identification of the cellular proteome in spatially confined areas around the POI. The key to *in situ* proximity tagging is the proximity tagging agent. Usually a small molecule or protein, the proximity tagging agent is responsible for activating an inert probe that contains an affinity handle and generating a reactive warhead that decorated nearby proteins with an affinity handle (e.g., biotin; Figure 1). The affinity handle replaces the need of a strong binding interaction between the POI and its interactors, as required by Y2H and AP-MS. Another advantage with proximity tagging is the expansion of captured targets beyond direct contact. Proximity labeling has created the opportunity to not only interrogate direct PPIs, but also the neighboring protein environment of a POI, enabling the identification of secondary interactors.

We were inspired by prior work with proximity tagging of GBPs (Siglec-2 and Siglec-15) where tagged carbohydrate recognition domains were fused with an antibody Fc domain. Used together with a secondary horse radish peroxidase (HRP)-conjugated anti-tag antibody, this system facilitated proximity tagging by tyramide signal amplification.[50] While this method successfully identified known interactors, we wanted to avoid using Fc domains, which could impart unnatural oligomerization. Furthermore, the sequential assembly of exogenous protein complexes for tagging would put a sizable distance between the source of activated proximity tagging probes (HRP) and the counter-receptors, necessitating longer labeling times and inducing more background labeling. In addition, the system was incapable of intracellular labeling, since HRP was known to have little to no activity inside live cells.[51]

At the time (2018), the two foremost technologies for proximity tagging were TurboID[52] and APEX2.[53] Both techniques enabled tagging in both extracellular and intracellular environments, and both protein domains are monomeric. These techniques did, however, differ in reaction kinetics. APEX2 was capable of a significant amount of labeling within 1 min, whereas TurboID required up to 10 min. The advantage of this rapid labeling would be the final piece of the puzzle necessary to begin investigations into glycoprotein-GBP interactions.

PROXIMITY TAGGING OF GBPS TO IDENTIFY NOVEL GLYCAN-GBP INTERACTIONS

We started our investigations by using GBPs as baits, because we would be able to easily add in proximity tagging agent-conjugated GBPs into cells to mimic the natural way GBPs interact with cell surface glycoconjugates. Galectin-3 (LGALS3) stood out as a prime GBP to begin our studies, because of several parameters. From a structural standpoint, its glycan-binding preferences have been well-annotated in glycan microarrays,[54,55] and synthetic potent inhibitors of LGALS3 have been described (e.g., TD139).[56] More importantly, LGALS3 was known to play diverse roles in development and disease,[57] and we hypothesized that its pleiotropic roles could be caused by variable engagement with different glycoprotein counter-receptors and their dynamic glycoforms. It was also notable that LGALS3 played both extracellular as well as intracellular roles. Extracellular LGALS3 had been implicated in trans-differentiation of stellate cells,[58] in apoptotic regulation of immune cells,[59] and in controlling dendritic cell differentiation,[60] whereas intracellular LGALS3 had been implicated in lysosome function in the cytosol[61] and in

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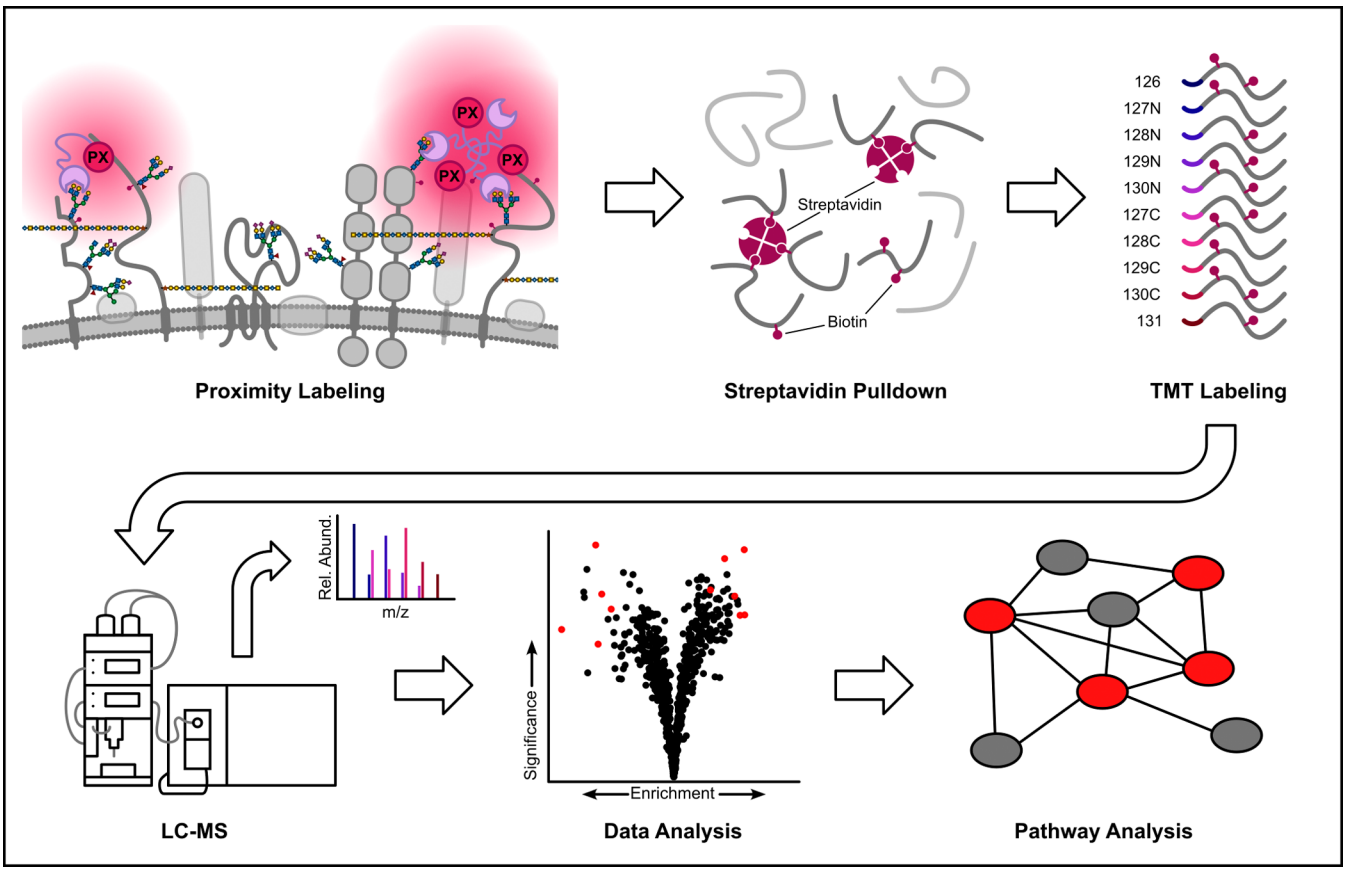


Figure 2. Schematic diagram of proximity tagging workflow. Proximity tagging experiments tag nearby proteins with biotin. The labeled proteins are pulled down with streptavidin, labeled with tandem mass tags (TMT), then sent through liquid chromatographic separation and detection by high-resolution mass spectrometry (LCMS) workflow. The resulting data of the relative abundances of TMT fragments is analyzed for the enrichment and statistical significance. This analyzed data can then be applied toward inferences to the interactome and functional roles of the GBP using bioinformatic databases. PX indicates the protein APEX2.

pre-mRNA splicing in the nucleus.[62] Furthermore, there was ample literature on its roles in assisting the immune response and immunosuppression in the tumor microenvironment. These diverse roles compelled us to investigate LGALS3 binding and dig deeper into missing gaps. While the glycan-binding preferences of LGALS3 were well understood through glycan arrays, additional preferences toward specific counterreceptors were not, with only a handful of known interactors at the time, including EGFR, TCR, and CEA.[57] Beyond this underexplored niche, it also appeared to us that LGALS3 could be uniquely poised to exemplify many of the key principles that proximity tagging would excel in demonstrating. LGALS3 is a monomeric protein that can oligomerize. It consists of a C-terminal carbohydrate recognition domain (CRD) (a feature shared across the galectins) and an N-terminal intrinsically disordered domain responsible for the oligomerization capacity of LGALS3.[63] In our initial experiments, we wanted to show that proximity tagging would not disturb the polyvalent activity of LGALS3 and preserve the endogenous binding of LGALS3 to its counter-receptors and strengthen the case for using proximity tagging to explore glycan-GBP interactions. In addition, with the difference in intracellular and extracellular activities of LGALS3, distinguishing extracellular versus intracellular interactomes would have high value.

In designing a proximity tagging approach to study LGALS3 (Figure 2), we first needed to construct a recombinant LGALS3 construct with a proximity tagging agent conjugated in a location that would minimally interfere with its glycan-binding activity. Fortunately, we were encouraged by prior work on LGALS3 fusion proteins, which showed that fusing protein tags onto the N-terminus (e.g., GFP) interfered minimally with LGALS3 binding properties.[64] Using APEX2 as our proximity tagging agent of choice, we conjugated APEX2 onto the N terminus of LGALS3 to generate recombinant APEX2-LGALS3 and confirmed that appending APEX2 caused minimal perturbations on the glycan-binding properties using ELISA with immobilized asialofetuin, a glycoprotein isolated from fetal calf serum, that displays the requisite galactose motifs. To further investigate the role of the N-terminal domain, we tested an additional construct lacking this domain. Based on ELISAs, we observed that the full-length APEX2-LGALS3 and the mutant lacking the N-terminal domain bound asialofetuin similarly (EC_{50} ~ 100 nM and 140 nM, respectively), suggesting that in such *in vitro* experiments, the oligomerization domain does not play a significant role in binding. The binding observed in ELISA was also glycan dependent as shown by apparent decreases in binding affinity in the presence of lactose. In live cells, proximity tagging was achieved at low concentrations between 25 and 100 nM. Notably, when using the full-length APEX2-LGALS3 and the N-terminal domain lacking mutant in live-cell proximity tagging, we noted a distinct decrease in labeling activity in the mutant, demonstrating that the N-terminal oligomerization domain was critical for maintaining the endogenous binding activity of the GBP in a cellular context. This observation exemplifies the

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important differences between an ELISA plate in comparison to the native environment of the cell surface, which in turn, emphasizes the importance of the oligomerization capacity of LGALS3.[1]

With our proximity tagging platform characterized and working, we began exploring the counterreceptors of LGALS3 by proteomics. Due to the density of the proteins on the cell surface and the nature of proximity tagging, we expected to tag both direct counter-receptors and interactome proteins, as well as proteins that interacted with LGALS3 in a nonglycan-mediated manner. Thus, careful consideration went into selecting controls for proximity tagging. We tested the use of tunicamycin, PNGAseF, TD139, and lactose − the first two inhibit N-glycosylation of cells, whereas the latter two inhibit the CRD of LGALS3. Using these controls, we identified 208 potential glycan-mediated binding partners of LGALS3 and validated a few of them. Leveraging the intracellular capabilities of APEX2, we also explored the intracellular LGALS3 interactome. Here, we identified 431 potential binding partners for LGALS3, and our profiles indicated that the intracellular interactome of LGALS3 is largely independent of its glycan-binding capacity.

Throughout these experiments, we used tandem-mass tag (TMT) proteomics.[65] In comparison to the other available methods of running comparative proteomics like SILAC,[66] TMT enabled much higher throughput and fidelity to the relative quantitation of peptides in our samples, and it was critical to reducing the amount of off-target hits in our data sets. To reduce the amount of background proteins identified in our experiments, we filtered proteins identified by Proteome Discoverer by thresholding the identified proteins by requiring a greater than 10× TMT enrichment ratio and two or more identifiable peptides per protein to validate the detection of the protein.[1]

In summary, our initial work on proximity tagging of GBP interactomes confirmed the importance of the identification of GBP counterreceptors in the endogenous cellular environment, established the effectiveness of properly controlled proximity tagging, and added to the body of work supporting the utility of TMT proteomics. We were able to identify the importance of the N-terminal intrinsically disordered oligomerization domain of LGALS3, as well as the GBP-counterreceptor interaction differences between an ELISA plate and the cell surface, emphasizing the importance of GBP oligomerization and using this knowledge to achieve a robust midnanomolar binding of LGALS3 on the cell surface.[1]

Following this initial work, we wanted to demonstrate the capabilities of proximity tagging for capturing interactomes of another GBP. We pursued galectin-1 (LGALS1), which shares significant sequence homology in the CRD of LGALS3; however LGALS1, like all other galectins, lacks the N-terminal disordered domain, forming dimers through a separate interaction.[67] As with LGALS3, we found that the dimerization capacity of LGALS1 was also critical to its binding to counter-receptors, however for LGALS1, this dimerization was further complicated by the oxidation state of an exposed cysteine. Fortuitously, our recombinant constructs with APEX2 dimerized at a similar potency with unconjugated LGALS1. In addition, the incubation time of the galectin on the cell surface also influences its labeling. Extended incubation of LGALS1 on cells resulted in internalization, which increased the nonspecific labeling of endolysosomal cell components due to the confining properties of cellular compartments. While this may be desired in some other context, for the purposes of our investigation of cell surface binding partners of LGALS1, reduced incubation times reduced background labeling.[68]

To expand our understanding of the cell surface proteome and to complement our GBP interactome studies, we also used a "baitless" proximity tagging approach to enrich for the cell surface proteome and identify both glycosylated and non-glycosylated proteins.[69] We reasoned that by localizing APEX2 near the cell surface, we could achieve general labeling of the cell surface. This was accomplished by dosing cells with GGGYC-conjugated cholesterol to incorporate the GGGYC tag onto the cell surface, then use an LPETG-tagged APEX2 and an engineered sortase enzyme to conjugate the LPETG and GGGYC tags, effectively localizing APEX2 onto the cell surface as a cholesterol-linked protein. Previously, we had identified two binding partners of LGALS3, basigin and neuroplastin, which were both known for facilitating the cell surface localization of monocarboxylate transporters (MCTs). We applied this method for capturing the cell surface proteome to find changes in enrichment of the cell surface proteome, comparing between vehicle treatment with basigin knockdown in a human hepatic stellate cell line. Through this platform, we were able to validate the efficacy of this method through identification of changes in the cell surface localization of previously annotated interactors MCT1 and MCT4. We saw potential in this comprehensive characterization of the cell surface proteome as a quality control data set, potentially useful for correlating against cell surface proximity tagging data sets to fish out interactors from general cell surface localized proteins.

■ ENGINEERING AND PROXIMITY TAGGING OF PROTEOGLYCANS TO IDENTIFY NOVEL GLYCAN-GBP INTERACTIONS

Because glycans on carrier proteins can engage different GBPs and template various macromolecular complexes, it was reasonable to assume that protein interactomes were also regulated by changes in glycosylation. However, technology that facilities such understanding does not yet exist. To address this, we developed a platform to engineer glycoproteins with precisely controlled glycosylation, enabling targeted investigation of glycosylation through proximity tagging.

As an initial system for this technique, we selected syndecan-1 (SDC1), a proteoglycan decorated with glycosaminoglycans (GAGs), specifically heparan sulfate (HS) and chondroitin sulfate (CS). The intrinsically disordered extracellular backbone minimizes the impact of manipulation, while the long glycan chains remain functionally intact despite modifications to proteoglycan linkages. By substituting a propargyl tyrosine at key glycosites, we used click-chemistry[70] to precisely attach defined GAGs onto SDC1. To display these on the cell surface, we replaced the transmembrane and intracellular domains with a polyhistidine tag, anchoring the engineered proteoglycan via cholesterol-PEG2000-NTA, which presented NTA on the cell surface for stable attachment (Figure 3). Conjugating APEX2 onto this engineered construct then enables proximity tagging of the defined glycoform. CD spectra were taken to ensure minimal perturbation to endogenous secondary structure organization of the engineered SDC1, with results indicating a minor increase to α helix character, which was expected due to the presence of alpha helices in APEX2. Intriguingly, our preliminary data suggests that SDC1 displays different interactomes following proximity tagging based on its glycosylation state and its cellular location.[2]

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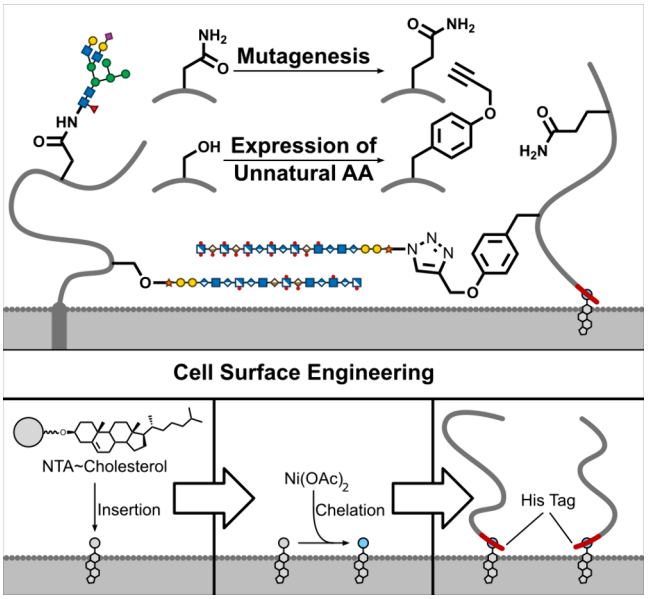


Figure 3. Schematic diagram of the methods of cell surface glycoform engineering. Upper Panel: Mutation of asparagine residues within canonical N-linked glycosylation sequons into glutamine residues removes glycosylation and expression of unnatural amino acids to remove canonical glycosylation sites and introduce click chemistry motifs, respectively. Lower Panel: lipid-anchored NiNTA anchors for recombinantly produced protein for introduction of defined glycoforms into the cell surface.

With this proteoglycan engineering platform established, we can begin to examine the roles of individual glycosites in other protein glycoconjugates like N-linked glycoproteins. There are now ongoing projects using proximity tagging investigating how the presence or absence of glycans on specified proteins alters interactomes (Figure 4), and discovering how protein glycoconjugates can template macromolecular complexes at the cell surface.

CHARACTERIZING CHANGING INTERACTOMES AND IDENTIFYING FUNCTIONAL GLYCOPROTEINS

As our proximity tagging techniques matured, we sought to uncover glycan-GBP interactions with direct functional implications, such as those occurring with the potent mitogen, fibroblast growth factor 2 (FGF2).[71] The interaction between FGF2 and its counter-receptor fibroblast growth factor receptor is mediated by a GAG, and this GAG is carried by a separate proteoglycan rather than the counter-receptor itself.[72] The proteoglycans in this context have remained understudied, and our proximity tagging platform was set perfectly to explore this space. We began by generating an APEX2-FGF2 construct and validating its binding efficacy and functionality through ELISA and a measurement of its ability to activate MAPK signaling cascades. Using heparin as a competitor to saturate the GAG-binding sites of FGF2, we distinguished protein versus GAG-mediated FGF2 interactors in A549 lung adenocarcinoma cells, identifying 147 GAG-mediated FGF2 interactors. Notably within the results, we found CD44 and neuropilin-1 (NRP1), both of which can be decorated by CS or HS, depending on cell type. Further validation was performed by Western blotting for shifts in apparent molecular weight caused by removal of these GAGs. We found that CD44 was not decorated with GAGs, and that NRP1 was a CS-decorated proteoglycan. Transient

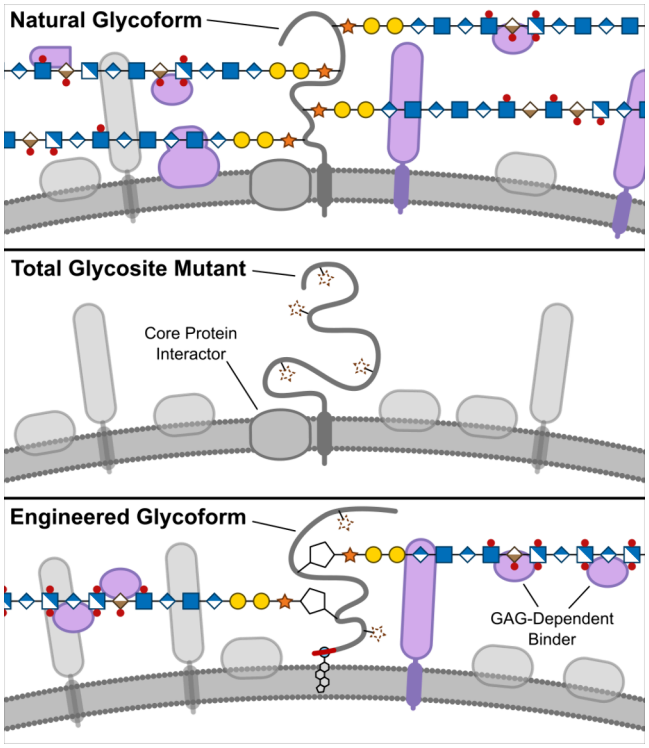


Figure 4. Schematic diagram of cell surface glycoform engineering and proximity tagging. By mutating glycosylation sites (glycosites) to eliminate protein glycosylation or to introduce engineered glycosylation, the roles of a glycan of interest at a particular glycosite can be examined by surveying the changing interactome.

knockdown of NRP1 alone was sufficient in reducing downstream phosphorylation and cell invasion that occurs in response to FGF2, identifying NRP1 as a critical GAG carrier protein in FGF2 signaling. It should be noted, however, that this proximity tagging experiment likely did not identify every relevant GAG-carrying proteoglycan. We have found that GAGs can effectively occlude biotinylation and shield the proteoglycan core from labeling and subsequent identification downstream. Further work to optimize these tagging events are underway.[73]

In addition to the work on FGF2 signaling, there are also ongoing projects in the lab exploring the functional roles of other glycoproteins. We are currently investigating function in proteins that originated from our results in the initial LGALS3 proximity tagging experiments, with CD9 and basigin. At the same time, we are also characterizing the shifts in the LGALS3 interactome associated with the changes between healthy and cancerous cell states.

OPPORTUNITIES TO ADVANCE PROXIMITY TAGGING FOR STUDYING GLYCAN-GBP INTERACTIONS

Our work in proximity tagging has demonstrated its utility in exploring the transient and easily perturbed interactions between glycoproteins and GBPs; however, we also note areas where improvements can be made to advance the precision and relevance of our results. One of our foremost considerations with proximity tagging is the differentiation of functional interactors from direct counter-receptors. Advancements in proximity tagging agents and ongoing improvements to the processing of proximity tagging results are enabling the further refinement of our understanding of glycan-GBP interactions.

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Among the possible improvements to basic characteristics of proximity tagging techniques, the reduction in labeling radius would still significantly enhance the precision of results obtained from proximity tagging. At the scales examined in proximity tagging, the labeling radius can be affected by the diffusion of activated probes from the proximity tagging agent, but also by the length of the linker to the catalytic center of the proximity tagging agent.[74] Thus, the enhancement of labeling radius can be improved on two fronts, reduction in the size and linker lengths of the proximity tagging agent, as well as improvement on the reactivity of the activated proximity tagging agent. In the case of examining glycan-GBP interactions, both improvements can in principle, be achieved by using small molecule photocatalysts.[74] By using the photocatalyst, the linker length can be controlled while the catalytic core would be directly connected onto the linker. Furthermore, these photocatalysts enable the use of much more reactive proximity tagging probes, minimizing half-lives of the reactive intermediates while taking advantage of the much more temporally controllable light-activated catalysis afforded by this technique.

In parallel with improving the proximity tagging agent is the incorporation of dual affinity purification techniques, such as ProLORT, which leverages the presence of lysine acetylation on their targets to orthogonally enrich targets preceding biotin enrichment.[75] Another similar example of dual affinity purification is in the isolation of mitochondria based on proximity to other organelles of interest. By affinity pulldown of mitochondria then biotin, the two orthogonal purifications greatly focus and reduce background pulldown of cellular material.[76] In a similar vein, the use of a split-APEX based technique may also refine the results of proximity tagging with the additional requirement of the association of two suspected interactors, greatly reducing the background labeling of APEX tags not localized to interaction sites.[77] While no analogous technique to these examples have been demonstrated for glycosylated proteins yet, improved purification techniques would enhance the selectivity toward glycosylated proteins beyond what biotinylation-based enrichment affords.

Ultimately, these above improvements work into the design of a proximity tagging experiment. The most critical factor to any proximity tagging experiment is the design of control experiments. A well-designed control can account for much of the proteins nonspecifically enriched by proximity tagging. The ability of a control condition to account for nonspecific enrichment is limited, however, in systems where the POI and proximity labeling agent are overexpressed. Differential expression of the proximity labeling agent as well as differential expression of proteins induced by the perturbations caused by the overexpression of the POI are both still challenging to control. We hypothesize that we may be able to use inducible-expression systems to titrate and control the expression of POI, while quantification of the abundance of proteins on the cell surface proteome may help to control for nonspecific enrichment. Otherwise, accumulation of proximity tagging data sets and the incorporation of machine learning tools like the random forest classifier recently demonstrated,[78] may also enhance our ability to differentiate between true interactors and bystander proteins in proximity tagging data sets.

■ CONCLUSIONS

Proximity tagging has created an exciting opportunity in glycoscience for the agnostic identification of glycoprotein-GBP interactions, and it promises to elucidate many interactions in live cells, where the presentation of binding partners is preserved. We are excited to see future expansion in the area, in terms of new classes of GBPs being explored and improved fidelity of putative interactors. In this Account, we explored the importance of preserving the endogenous activity of GBPs and presentation of their counterreceptors. Our work further emphasizes the fidelity that proximity tagging techniques can bring to the study of such transient and dynamic interactions. We also have demonstrated that the results of proximity tagging experiments lend themselves to functional glycobiological explorations of the role glycosylation plays in the cell surface proteome.

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CRediT: **Patrick Tseng** writing - original draft, writing - review & editing. Mia L. Huang - conceptualization, review & editing.

Notes

The authors declare no competing financial interest.

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Patrick Tseng is a Ph.D. student in the laboratory of Mia L. Huang at Scripps Research. He earned his B.S. in Biochemistry, magna cum laude, from the University of California, San Diego in 2023. His current research focuses on implementing novel methodologies for proximity tagging of glycoconjugates.

Prof. Mia L. Huang is a Professor in the Department of Chemistry at Scripps Research. She earned her B.A. in Chemistry from CUNY Queens College in 2007 and her Ph.D. in Chemistry from New York University in 2012 with Prof. Kent Kirshenbaum. She carried out postdoctoral research focusing on chemical glycobiology at UC San Diego with Prof. Kamil Godula and joined the Scripps Research faculty in 2018. Her group focuses on research into chemical methods of studying glycoconjugates.

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